SYLLABUS FOR B. TECH
COMPUTER SCIENCE AND ENGINEERING COURSE
GAUHATI UNIVERSITY

Abbreviations Used
AS – Applied Science Course
IT – Information Technology Course
ECE – Electronics and Communication Engineering Course
BT – Bioengineering and Technology Course
ESC – General Engineering
HS – Humanities and Social Science
MG – Management Course

Course code Structure for Core Papers

XXABC means: XX for Department/Nature of the course
Digit A for year
Digit B for semester (1 for odd semester and 2 for even semester)
Digit C for serial number of the course

Course code Structure for Elective Papers
XX9YY means: XX for Department/Nature of the course
YY for Serial number of the elective course

SYLLABUS STRUCTUTE

Semester 1								
SL	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	AS114	Physics	3	1	1.5	5.5	7	
2	AS113	Mathematics-I	3	1	0	4	4	
3	ECE111	Basic Electrical Electronics Engineering	3	1	1	5	6	
4	ESC111	Engineering Graphics & Design	1	0	2	3	5	
5	IT111	Fundamentals Of Computing	2	0	1	3	4	
	IND111	Induction Programme	0	0	0	0	0	Audit Course
		Total	10	3	4.5	20.5	26	

Semester 2								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	AS121	Chemistry	3	1	1.5	5.5	7	
2	AS122	Mathematics-II	3	1	0	4	4	
3	IT121	Programming for Problem Solving	2	0	1	3	4	
4	ESC121	Workshop	1	0	2	3	5	
5	HS121	English	2	0	1	3	4	
6	BT121	Modern Biology	2	0	1	3	4	
7	VAC121	YOGA	0	0	2	0	4	VAC Course
	VAC122	NCC	0	0	2	0	4	
	VAC123	NSS	0	0	2	0	4	
		Total	12	2	6.5	21.5	28	

Semester 3								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	IT211	Data Structure And Algorithm	3	0	1	4	5	For CSE, IT, BET
2	IT212	It Workshop	0	0	3	3	6	For CSE, IT
3	ECE211	Digital Electronics	3	0	1	4	5	
4	AS212	Probability And Statistics	3	1	0	4	4	
5	HS211	English 1	3	0	0	3	3	Select Any One
	HS212	Elements Of Economics						
6	INT211	Internship	-	-	-	4	0	Carried from semester II
		Total				22	23	

Semester 4								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	IT221	Computer Organization and Architecture	3	0	1	4	5	
2	IT222	Object Oriented Programming	3	0	1	4	5	
3	IT223	Discrete Mathematics	3	1	0	4	4	
4	IT224	Data Communication & Computer Network	3	0	1	4	5	
5	MG 221	Management 1 (Finance & Accounting)	3	0	0	3	3	
6	HS 221	Environmental Sciences				0	2	Audit Course
		Total				19	24	

Semester 5								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	IT311	Database Management Systems	3	0	1	4	5	
2	IT312	Design and Analysis of Algorithms	3	1	0	4	4	
3	IT313	Operating Systems	3	0	1	4	5	
4	IT314	Software Engineering	3	1	0	4	4	
5	IT315	Advanced Programming	1	1	1	3	4	
6	HS 311	Humanities-II (Organizational Behaviour)	3	0	0	3	3	
7	HS 312	Constitution of India				0	2	Audit Course
	HS 313	Essence of Indian Knowledge Tradition						
8	INT 311	Internship				5	0	Carried from semester IV
		Total				27	27	

Semester 6								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	IT321	Formal Language and Automata Theory	3	0	1	4	5	
2	IT322	Artificial Intelligence & Machine Learning	3	0	1	4	5	
3	IT323	Python Programming	3	0	1	4	5	
4		Dept. Elective-I				3	3	
5	IT323	Development Project	-	-	2	2	0	
6	HS 32X	Open Elective-I	3	0	0	3	3	Humanities
		Total				20	21	

Semester 7								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	IT411	Cyber Security	3	0	1	4	5	
2		Dept. Elective-II				3	3	
3		Dept. Elective-III				3	3	
4		Open Elective-II				3	3	
5		Science Elective				3	3	
6	IT412	Project				6	0	
7	INT411	Internship				5	0	Carried from semester VI
		Total				23	17	

Semester 8								
Sl	**Course Code**	**Course Title**	**L**	**T**	**P**	**C**	**Contact hrs**	**Remarks**
1	IT421	Project	-	-	-	20		
2		Mooc Course						Audit Course
		Total				20		

List of Professional Electives for BTech CSE								
Data Science and Machine Intelligence								
1	IT901	Deep Learning						
2	IT902	Speech Processing						
3	IT903	Neural Network and Machine Learning						
4	IT904	Natural Language Processing						
5	IT905	Nature Inspired Computing						
6	IT907	Mathematics for Machine Learning						
7	IT908	Data Mining & Warehousing						
Development								
1	IT910	C# Programming						
2	IT911	Web Technology						
3	IT912	Human Computer Interfacing						
4	IT913	Image Processing						
Systems								
1	IT920	Advance Computer Organization and Architecture						
2	IT921	System Programming						
3	IT922	Embedded System						
4	IT923	Distributed system						
5	IT924	System Administration						
6	IT925	Internet of Things						
Networks								
1	IT930	Network Programming						
2	IT931	Wireless Network						
3	IT932	Adhoc and Sensor Network						
4	IT933	Cryptography and Network Security						
Theory and Algorithms								
1	IT940	Graph Theory						
2	IT941	Theory of Computation						
3	IT942	Programming Language						
4	IT943	Information Theory and Coding						
5	IT944	Operation Research						
6	IT945	Compiler Design						
List of Open Electives for BTech CSE								
1	IT950	Software Testing, Quality and Security Management						
2	IT951	Spread sheet Analysis & Programming						
3	IT952	Mobile Application Development						
4	IT953	Programming with R						
5	IT954	Optimization Techniques						